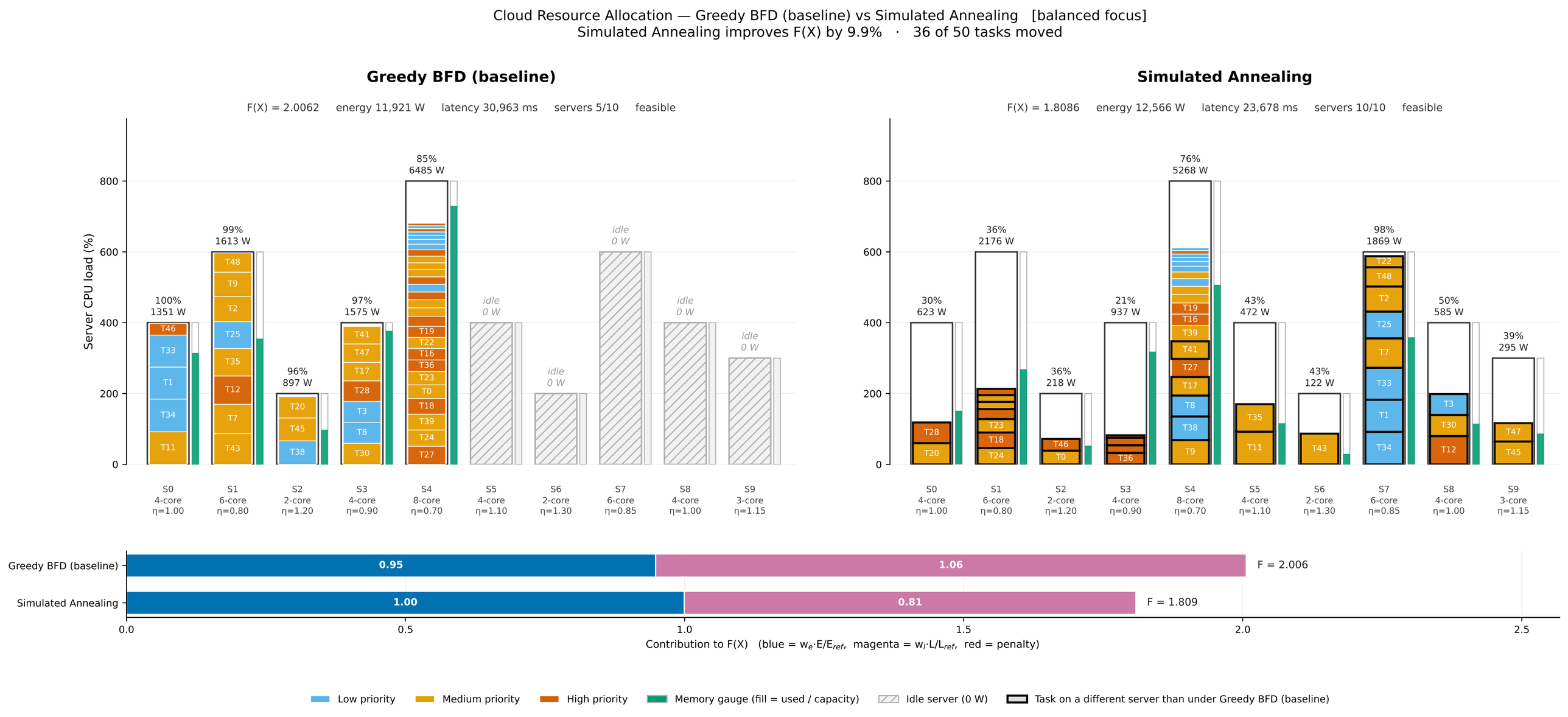
Cloud Resource Allocation — Greedy BFD (baseline) vs Simulated Annealing [balanced focus]
Simulated Annealing improves F(X) by 9.9% · 36 of 50 tasks moved
Greedy BFD (baseline)
F(X) = 2.0062 energy 11,921 W latency 30,963 ms servers 5/10 feasible
Simulated Annealing
F(X) = 1.8086 energy 12,566 W latency 23,678 ms servers 10/10 feasible
Server CPU load (%)
100% 1351 W
99% 1613 W
96% 897 W
97% 1575 W
85% 6485 W
idle 0 W
30% 623 W
36% 2176 W
36% 218 W
21% 937 W
76% 5268 W
43% 472 W
43% 122 W
98% 1869 W
50% 585 W
39% 295 W
S0 4-core η=1.00
S1 6-core η=0.80
S2 2-core η=1.20
S3 4-core η=0.90
S4 8-core η=0.70
S5 4-core η=1.10
S6 2-core η=1.30
S7 6-core η=0.85
S8 4-core η=1.00
S9 3-core η=1.15
Greedy BFD (baseline)
Simulated Annealing
0.95
1.06
F = 2.006
1.00
0.81
F = 1.809
Contribution to F(X) (blue = $w_e \cdot E/E_{ref}$, magenta = $w_l \cdot L/L_{ref}$, red = penalty)
Low priority
Medium priority
High priority
Memory gauge (fill = used / capacity)
Idle server (0 W)
Task on a different server than under Greedy BFD (baseline)